

$$\sum \frac{1}{d-p_i} = \sum \frac{1}{d-z_j}$$

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graph TD; A["\sum \frac{1}{d-p_i} = \sum \frac{1}{d-z_j}"] --> B["necessary condition only"]; B --> C["candidate point must also lie on root locus"]; C --> D["there may be several breakaway points"]; D --> E["breakaway point is not always on the real axis"];
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necessary condition only

candidate point must also lie on root locus

there may be several breakaway points

breakaway point is not always on the real axis